

PFAS results

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PFAS

Category and filtration

```
Family: Gamma ( log )
Formula:      concentration ~ category
Zero inflation: ~.
Data: pfas.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
523.7	533.5	-256.9	513.7	47

Dispersion estimate for Gamma family (sigma^2): 0.368

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.5472	0.1264	43.88	< 2e-16 ***
categoryMunicipal	0.9889	0.2223	4.45	8.61e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.3023	0.3198	-0.945	0.3446
categoryMunicipal	-2.0956	1.0923	-1.919	0.0551 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Confidence intervals

Gamma components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	200.22	328.65	256.52
cond.categoryMunicipal	1.74	4.16	2.69

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	2.53	0.72	1.35
zi.categoryMunicipal	69.17	0.96	8.13

Conclusions

```
pvals.pfas <- NULL
# Conclusion 1
# category, zero inflated component
pvals.pfas[1] <- 0.0551
# category, conditional component
pvals.pfas[2] <- 8.61e-06
# Conclusion 2
# source, fisher pitman
pvals.pfas[3] <- 2.93e-08
# Conclusion 3
# advanced filtration, fisher pitman
pvals.pfas[4] <- 0.004812
names(pvals.pfas) <- c(
  "Conclusion.pfas1.1", "Conclusion.pfas1.2",
  "Conclusion.pfas2",
```

```
"Conclusion.pfas3"
)
```

Conclusion.pfas1.1	Conclusion.pfas1.2	Conclusion.pfas2	Conclusion.pfas3
0.06	0.00	0.00	0.01

PFAS Writeup

Conclusion 1: Bottled v. municipal

Conclusion 1:

- The odds of any PFAS being detected in a municipal water sample is 8.13 times higher compared to bottled water (adjusted ratio: 8.13, 95% interval: [0.96, 69.17]), keeping advanced filtration fixed. This confidence interval is quite wide.
 - Unadjusted p-value: 0.0278
 - Adjusted p-value: 0.06
- With 57% of bottled samples having PFAS detected, the probability of detection for municipal water is 1.59 higher than in bottled water (adjusted ratio: 1.59, 95% interval for relative risk: [0.98, 1.72]).
 - Unadjusted p-value: 0.0278
 - Adjusted p-value: 0.06
- For samples with PFAS detected, the mean PFAS concentration in a municipal water sample is 2.69 times higher compared to bottled water (adjusted ratio: 2.69, 95% interval: [1.74, 4.16])).
 - Unadjusted p-value: 0.000835
 - Adjusted p-value: 0
- Overall, the evidence suggests municipal water has more PFAS compared to bottled water.

Conclusion 2: Spring v. non-spring

Conclusion 2:

- We want to compare spring and non-spring sources. However, this is a challenging task. There are no municipal sources from spring water. Thus, if I tell you a sample is spring water, you know it is definitely bottled water. This makes it hard to disentangle the effect of the source from whether a sample is from bottled or municipal water. Thus, I want

to restrict to looking at spring v. non-spring source for bottled water only. However, this still remains challenging—among non-spring bottled water, all of them have no PFAS detected! So, I should not be looking at mean concentration comparing spring and non-spring sources, because so many samples have 0 concentration. Thus, we only compare whether the proportion of samples with PFAS detected is different among spring and non-spring sources among bottled water.

- Among bottled water samples, the proportion of samples with PFAS detected is significantly higher in spring sources (89%, 23/26) compared to non-spring sources (0%, 0/14).
 - Unadjusted p-value: 2.93e-08
 - Adjusted p-value: 0
- There is strong evidence that in bottled water, spring sources are more likely to have PFAS detected.

Conclusion 3, advanced filtration

- We have similar problem as for source type. All samples with advanced filtration had 0 PFAS detected. We have the additional problem that source type and advanced filtration are highly collinear—all advanced filtration samples in bottled water come from non-spring sources. So, for both these reasons, again a zero-inflated model does not make sense, and instead we again apply a Fisher exact test.

Conclusion 3:

- Among bottled water samples, the proportion of samples with PFAS detected is significantly higher without advanced filtration (72%, 23/32) compared to with advanced filtration sources (0%, 0/8).
 - Unadjusted p-value: 0.004812
 - Adjusted p-value: 0.01
- There is strong evidence that in bottled water, advanced filtration is less likely to have PFAS detected.